Lecture 6: MAP Estimation

Priors · Posteriors · Regularization Connection

Previously, on Lecture 5...

MoM: Set population moments = sample moments. Simple but can give impossible values.

MLE: $\hat{\theta} = \arg\max \ell(\theta)$. Pick the θ that makes the data most probable.

Properties: Consistent, asymptotically normal, efficient, invariant.

MLE = ML: Gaussian noise $\rightarrow$ MSE loss. Bernoulli $\rightarrow$ cross-entropy.

But: MLE can overfit with small n or flexible models.

Today: What if we have **prior knowledge** about θ?

Can we do better than MLE by incorporating beliefs *before* seeing data?

Where We Are

MLE
maximize $P(\text{data} \mid \theta)$

What if we have prior beliefs?

MAP
maximize $P(\theta \mid \text{data})$

Lecture 5

This lecture

Bayes' Theorem for Parameters

From likelihood to posterior

Bayes' Theorem for Parameters

$$\underbrace{P(\theta \mid \text{data})}_{\text{posterior}} = \frac{\overbrace{P(\text{data} \mid \theta)}^{\text{likelihood}} \cdot \overbrace{P(\theta)}^{\text{prior}}}{\underbrace{P(\text{data})}_{\text{evidence}}}$$

Bayes' Theorem for Parameters

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Or simply: posterior $\propto$ likelihood $\times$ prior

The Three Ingredients

Prior $P(\theta)$

What you believed *before* seeing data

"Coins are usually close to fair"

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Likelihood $P(\textbf{data} \mid \theta)$

What the data says (same as MLE)

"7 heads in 10 flips"

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$\times$

Likelihood $P(\textbf{data} \mid \theta)$

What the data says (same as MLE)

"7 heads in 10 flips"

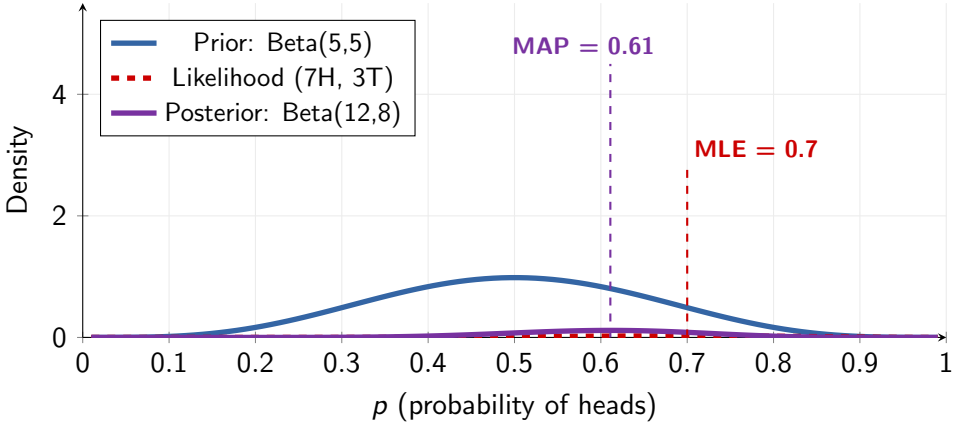
$=$

Posterior $P(\theta \mid \textbf{data})$

Updated belief *after* seeing data

"Coin is probably slightly biased"

Visualizing the Update: Coin Bias



Prior pulls the estimate from 0.7 toward 0.5. The posterior is a **compromise**.

Conjugate Priors as Pseudo-Observations

Why did the math work out so neatly? Because Beta is **conjugate** to Binomial:

Prior: $\text{Beta}(\alpha, \beta)$ $+$ Data: k H, $n-k$ T $=$ Posterior: $\text{Beta}(\alpha+k,\ \beta+n-k)$

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Prior: $\text{Beta}(\alpha, \beta)$ + Data: k H, $n-k$ T = Posterior: $\text{Beta}(\alpha+k,\ \beta+n-k)$

The prior acts like "fake data" you've already seen:

$\text{Beta}(\alpha, \beta)$ = pretend you already observed $\alpha-1$ heads and $\beta-1$ tails.

$\text{Beta}(5, 5)$: "I've seen 4H and 4T" (8 pseudo-observations).

After 7H, 3T (10 real obs): posterior = $\text{Beta}(12, 8)$ = "11H, 7T out of 18 total."

As $n \to \infty$, the pseudo-observations become negligible $\Rightarrow$ posterior $\to$ likelihood $\Rightarrow$ MAP $\to$ MLE.

Common Conjugate Pairs

Likelihood	**Conjugate Prior**	**Posterior**	**Pseudo-data**
Binomial(n, p)	Beta(α, β)	Beta$(\alpha+k,\ \beta+n-k)$	$\alpha-1$ H, $\beta-1$ T
Poisson(λ)	Gamma(a, b)	Gamma$(a+\sum x_i,\ b+n)$	a events in b time
$N(\mu, \sigma_0^2)$	$N(m, \tau^2)$	$N\left(\frac{n\bar{x}/\sigma_0^2+m/\tau^2}{n/\sigma_0^2+1/\tau^2},\ \ldots\right)$	1 obs of value m
Exp(λ)	Gamma(a, b)	Gamma$(a+n,\ b+\sum x_i)$	a events in b time

Pattern: if the likelihood is in the exponential family, there is always a conjugate prior.
The posterior update just adds the sufficient statistics.

Bayesian Updating Is Sequential

A key Bayesian insight: **today's posterior becomes tomorrow's prior**.

$$\underbrace{\text{Prior } \mathrm{Beta}(1,1) \xrightarrow{+\,\text{data}} 3\text{H, } 1\text{T} \xrightarrow{=} \mathrm{Beta}(4,2)}_{\text{Day 1}} \underbrace{\xrightarrow{+\,\text{data}} 5\text{H, } 2\text{T} \xrightarrow{=} \mathrm{Beta}(9,4)}_{\text{Day 2}}$$

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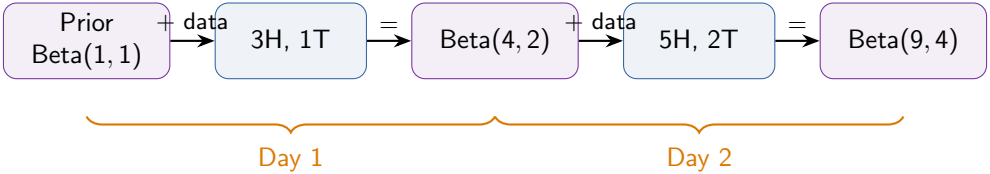

Order doesn't matter! Processing all 8H, 3T at once gives the same Beta(9, 4).

Bayesian updating is **consistent** — you can pause and resume at any time.

MAP Estimation

The mode of the posterior

MAP = Mode of the Posterior

$$\hat{\theta}_{\text{MAP}} = \arg\max_\theta \; P(\theta \mid \text{data}) = \arg\max_\theta \left[\ell(\theta) + \log P(\theta)\right]$$

Maximize: log-likelihood + log-prior

MLE: $\arg\max_\theta \; \ell(\theta)$ $\xrightarrow{+\text{ prior}}$ MAP: $\arg\max_\theta \; \ell(\theta) + \log P(\theta)$

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MAP = MLE with an extra penalty/bonus term from the prior.

MAP Step by Step: Coin Bias

Setup: Prior $p \sim \text{Beta}(5, 5)$, data: 7 heads in 10 flips.

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Step 1: Write the posterior.

$$P(p \mid \text{data}) \propto \underbrace{p^7(1-p)^3}_{\text{likelihood}} \cdot \underbrace{p^4(1-p)^4}_{\text{Beta(5,5) prior}} = p^{11}(1-p)^7$$

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Step 2: Find the mode.

$$\hat{p}_{\text{MAP}} = \frac{\alpha' - 1}{\alpha' + \beta' - 2} = \frac{12 - 1}{12 + 8 - 2} = \frac{11}{18} \approx 0.611$$

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Compare with MLE: $\hat{p}_{\text{MLE}} = \frac{7}{10} = 0.700$

The prior pulled the estimate from 0.70 toward 0.50 (the prior mean), landing at 0.61.
With more data, this effect shrinks.

MAP for a Normal Mean

Model: $X_1, \ldots, X_n \sim N(\mu, \sigma_0^2)$ (known σ_0^2), prior $\mu \sim N(m, \tau^2)$.

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$$\log P(\mu \mid \text{data}) = \underbrace{-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(X_i - \mu)^2}_{\text{log-likelihood}} + \underbrace{\left(-\frac{(\mu - m)^2}{2\tau^2}\right)}_{\text{log-prior}} + \text{const}$$

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Solve:

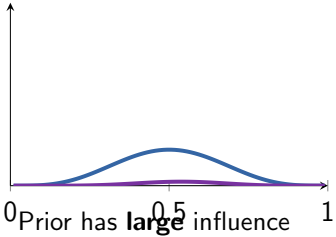
$$\boxed{\hat{\mu}_{\text{MAP}} = \frac{\frac{n}{\sigma_0^2}}{\frac{n}{\sigma_0^2} + \frac{1}{\tau^2}} \cdot \bar{X} + \frac{\frac{1}{\tau^2}}{\frac{n}{\sigma_0^2} + \frac{1}{\tau^2}} \cdot m = w \cdot \bar{X} + (1 - w) \cdot m}$$

A **weighted average** of the sample mean $\bar{X}$ and the prior mean m.

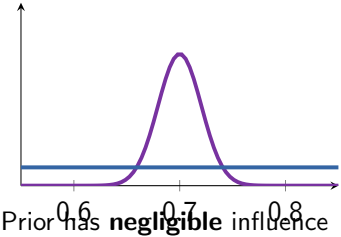
More data ($n \uparrow$) $\Rightarrow w \to 1 \Rightarrow$ MAP $\to$ MLE. Stronger prior ($\tau^2 \downarrow$) $\Rightarrow w \to 0 \Rightarrow$ MAP $\to m$.

When Does the Prior Matter?

Small n (e.g., $n = 5$)



Large n (e.g., $n = 500$)



With enough data, the likelihood dominates $\Rightarrow$ MAP $\approx$ MLE.
The prior is "washed out" by the data.

Choosing the Prior

Informative prior

Encodes real knowledge:
"Coins are close to fair"
$\rightarrow$ Beta$(50, 50)$

Stronger pull toward prior belief.
Fewer data needed.

Non-informative priors: Beta$(1, 1)$ = Uniform. Sounds "objective" but can be
problematic — the result depends on how you parameterize!
Jeffreys prior $P(\theta) \propto \sqrt{I(\theta)}$ is invariant to reparameterization (uses Fisher info from Lecture 4).

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Regularization as MAP

Priors are penalties in disguise

The Key Connection: Regularization = MAP

$$\text{MAP:} \quad \hat{\theta} \;=\; \arg\max_\theta \big[\ell(\theta) \;+\; \log P(\theta)\big]$$

is the same as

$$\text{Regularization:} \quad \hat{\theta} \;=\; \arg\min_\theta \big[-\ell(\theta) \;+\; \lambda \cdot \text{penalty}(\theta)\big]$$

The log-prior acts as a **penalty on the parameters**.

From Prior to Penalty: The Algebra

Assume: $y_i = \mathbf{x}_i^\top \theta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2), \quad \text{prior } \theta_j \sim N(0, \tau^2).$

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Combine and flip sign (max $\to$ min, drop constants):

$$\hat{\theta}_{\text{MAP}} = \arg\min_\theta \left[\underbrace{\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \theta)^2}_{\text{data fit (= MSE loss)}} + \underbrace{\frac{1}{2\tau^2} \|\theta\|_2^2}_{\text{from the prior}} \right]$$

Set $\lambda = \sigma^2/\tau^2$: **small** τ^2 (tight prior) $\Rightarrow$ **large** λ (strong penalty).
This is exactly **Ridge regression**.

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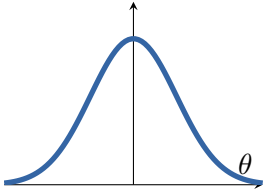
Gaussian Prior $\Leftrightarrow$ Ridge (L2) Regression

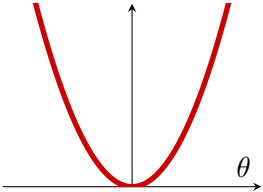
Gaussian prior



$P(\theta) = \mathcal{N}(0, \tau^2)$

$\Longrightarrow$

L2 penalty



$-\log P(\theta) \propto \|\theta\|_2^2$

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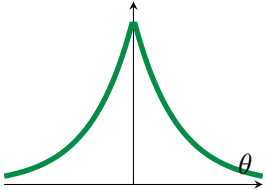
This is exactly **Ridge regression**! $\lambda = \sigma^2/\tau^2$

Laplace Prior $\Leftrightarrow$ Lasso (L1) Regression

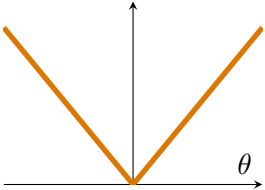
Laplace prior



$P(\theta) \propto e^{-|\theta|/b}$

$\Longrightarrow$

L1 penalty



$-\log P(\theta) \propto \|\theta\|_1$

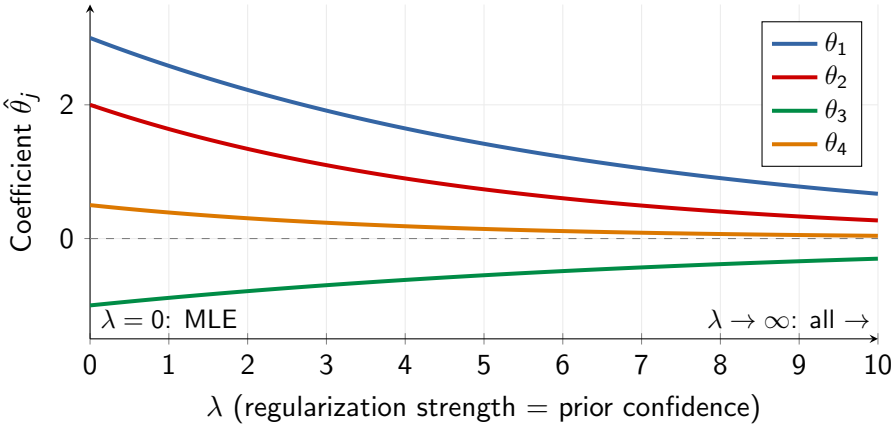
$$\hat{\theta}_{\text{MAP}} \quad = \quad \arg\min_\theta \left[\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \theta)^2 + \lambda \|\theta\|_1\right]$$

This is exactly **Lasso regression**! Encourages **sparse** solutions ($\theta_j = 0$).

The Regularization Map

Prior (Bayesian)	Penalty (Frequentist)
Gaussian $\mathcal{N}(0, \tau^2)$	Ridge: $\lambda\|\theta\|_2^2$ *shrinks toward 0*
Laplace$(0, b)$	Lasso: $\lambda\|\theta\|_1$ *sets some to exactly 0*
Gaussian + Laplace	Elastic Net: $\lambda_1\|\theta\|_1 + \lambda_2\|\theta\|_2^2$ *sparse + grouped shrinkage*
Uniform (flat)	No penalty *= MLE, no shrinkage*

Visualizing Ridge Shrinkage



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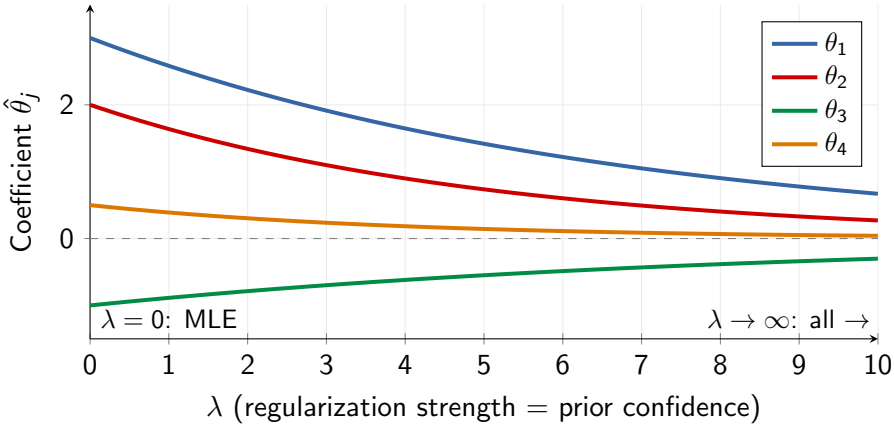
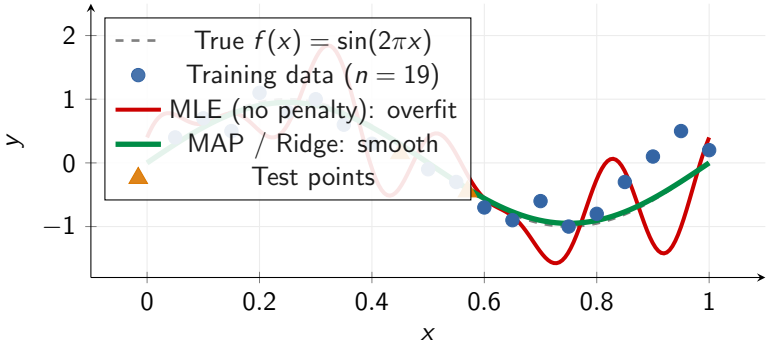


Increasing λ = stronger prior = more shrinkage = less overfitting (but more bias).

MLE vs MAP: The Overfitting Story

Fit a degree-15 polynomial to noisy data. MLE uses all coefficients freely; MAP (Ridge) penalizes large ones.



MLE:
Train: low
Test: high

MAP:
Train: medium
Test: low

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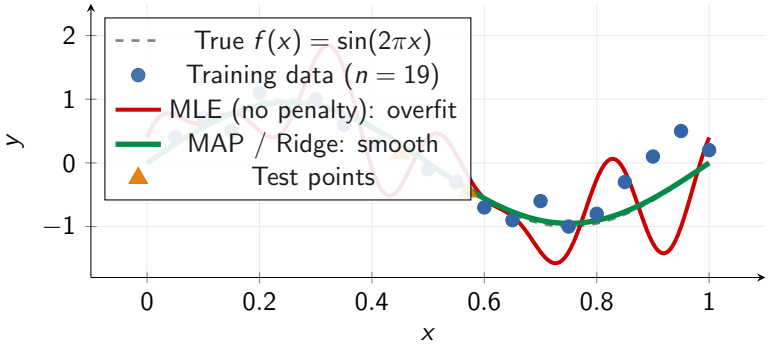

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The prior says "coefficients should be small" $\Rightarrow$ smoother fit $\Rightarrow$ better **generalization**.

MLE vs MAP vs Full Bayesian

Three philosophies of estimation

Three Philosophies

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Mode of likelihood

No prior info used

Risk: overfitting with small n

$\hat{\theta} = \arg\max \ell(\theta)$

$\xrightarrow{\text{add prior}}$

MAP

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Uses prior beliefs

Still a point estimate

$\hat{\theta} = \arg\max[\ell + \log P]$

$\xrightarrow{\text{keep full dist.}}$

Full Bayesian

Entire posterior

Full uncertainty

Computationally expensive (MCMC)

$P(\theta \mid \text{data})$

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Empirical Bayes (a hybrid): estimate the prior *from the data itself*. Use MLE to pick hyperparameters $(\alpha, \beta, \tau^2, \ldots)$, then do Bayesian inference.

Frequentist spirit + Bayesian machinery. Popular in genomics, NLP, and James–Stein shrinkage.

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– Large n (prior doesn't matter)
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– Want a point estimate fast

Full Bayesian when:
– Uncertainty quantification matters (medical, safety)
– Model comparison needed
– Computational cost is acceptable

Summary: From MLE to MAP

Bayes' rule: posterior $\propto$ likelihood $\times$ prior. Update beliefs with data.

Conjugacy: Beta–Binomial, Normal–Normal. Prior acts like pseudo-observations.

MAP: $\hat{\theta} = \arg\max[\ell(\theta) + \log P(\theta)]$. MLE with a prior bonus.

Normal MAP: Weighted average $w\bar{X} + (1-w)m$. More data $\Rightarrow$ MAP $\to$ MLE.

Gaussian prior $\to$ Ridge: $-\log P(\theta) \propto \|\theta\|_2^2$. Shrinks toward 0.

Laplace prior $\to$ Lasso: $-\log P(\theta) \propto \|\theta\|_1$. Drives coefficients to exactly 0.

Flat prior $\to$ no penalty: MAP reduces to MLE. No prior = no regularization.

Full Bayesian: Use the entire posterior, not just its mode. Gold standard but expensive.

Practical: Priors and Posteriors

1. **Coin bias estimation:**
   - Start with Beta(1,1), Beta(5,5), Beta(50,50) priors
   - Observe 7 heads in 10 flips
   - Plot prior, likelihood, and posterior for each
   - Compare the MAP estimates — how much does the prior pull?
2. **Ridge regression as MAP:**
   - Fit linear regression with $\lambda = 0, 0.1, 1, 10, 100$
   - Plot coefficients vs λ (shrinkage path)
   - Observe: larger λ = stronger prior = more shrinkage
3. **Visualize:** Plot the prior/likelihood/posterior for a simple 1D Normal with known σ^2, varying the prior variance τ^2

Homework

1. For $X_1, \ldots, X_n \sim N(\mu, \sigma_0^2)$ (known σ_0^2) with prior $\mu \sim N(m, \tau^2)$:
   derive the MAP estimator $\hat{\mu}_{\text{MAP}}$. Show it is a weighted average of $\bar{X}$ and m.
   What happens as $\tau^2 \to \infty$? As $n \to \infty$?

2. A coin is flipped 20 times with 14 heads. Compute the MAP estimate of p under:
   (a) Beta$(1, 1)$, (b) Beta$(5, 5)$, (c) Beta$(50, 50)$ priors.
   Compare with the MLE. Which prior has the most influence?

3. Show that Ridge regression $\hat{\theta} = \arg\min \left[\|y - X\theta\|^2 + \lambda \|\theta\|^2 \right]$
   has the closed-form solution $\hat{\theta} = (X^\top X + \lambda I)^{-1} X^\top y$.
   Why does this always have a unique solution, even when $X^\top X$ is singular?

Recommended Visualizations & Resources

Interactive: Bayesian Updating (R Psychologist)

`rpsychologist.com/d3/bayes` — drag sliders to adjust the prior and sample size; watch the posterior update in real time. The best interactive Bayesian demo.

Interactive: Seeing Theory — Bayesian Inference (Brown)

`seeing-theory.brown.edu/bayesian-inference` — drag priors to see posteriors update live. Beautiful animations for prior×likelihood=posterior.

Video: 3Blue1Brown — Bayes' Theorem

`3blue1brown.com/lessons/bayes-theorem` — stunning animated walkthrough of how prior beliefs transform into posteriors. The geometric intuition directly maps to MAP.

Interactive: Ridge & Lasso Visualization

`xavierbourretsicotte.github.io/ridge_lasso_visual.html` — contour plots of loss + penalty. See geometrically why Lasso gives sparse solutions.

Video: StatQuest — Bayes' Theorem, clearly explained

`statquest.org/bayes-theorem-clearly-explained` — step-by-step walkthrough of prior, likelihood, posterior with clear examples.

Questions?

Next: Lecture 7 — Sampling distributions and confidence intervals